Teoria da Viga de Euller-Bernoulli[1]

Euler–Bournoulli Beam Theory

Figure 6.10 illustrates a cantilevered beam with the transverse direction of vibration indicated [i.e., the deflection, $w(x, t)$, is in the y direction]. The beam is of rectangular cross section $A(x)$ with width h_y, thickness h_z, and length l. Also associated with the beam is a flexural (bending) stiffness $EI(x)$, where E is Young's elastic modulus for the beam and $I(x)$ is the cross-sectional area moment of inertia about the "z axis." From mechanics of materials, the beam sustains a bending moment $M(x, t)$, which is related to the beam deflection, or bending deformation, $w(x, t)$, by

$$M(x,t) = EI(x)\frac{\partial^2 w(x,t)}{\partial x^2} \tag{6.85}$$

A model of bending vibration may be derived from examining the force diagram of an infinitesimal element of the beam as indicated in Figure 6.10. Assuming the deformation to be small enough such that the shear deformation is much smaller than

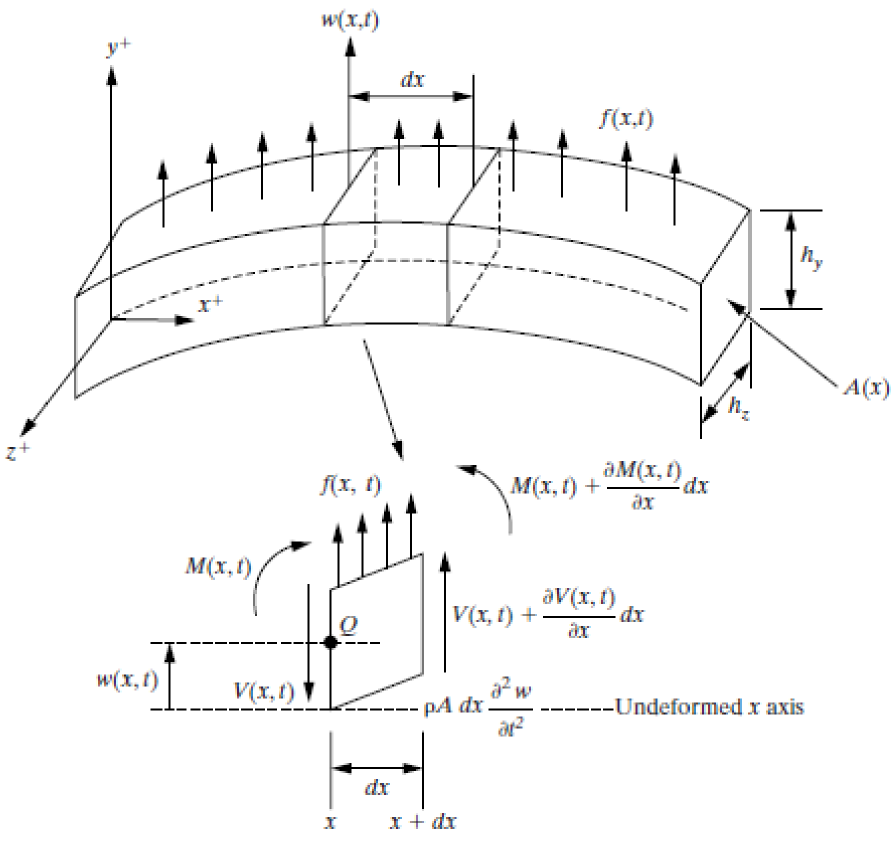


Figure 6.10 A simple Euler–Bernoulli beam of length (l) in transverse vibration and a free-body diagram of a small element of the beam as it is deformed by a distributed force-per-unit length, denoted by $f(x, t)$.

[1] Inman, D.J., Engineering Vibration, Pearson Ed., New York, 4ed., 2014.

$w(x, t)$ (i.e., so that the sides of the element dx do not bend), a summation of forces in the y direction yields

$$\left(V(x,t) + \frac{\partial V(x,t)}{\partial x} dx \right) - V(x,t) + f(x,t)dx = \rho A(x)\, dx \frac{\partial^2 w(x,t)}{\partial t^2} \quad (6.86)$$

Here $V(x, t)$ is the shear force at the left end of the element dx, $V(x, t) + V_x(x, t)\,dx$ is the shear force at the right end of the element dx, $f(x, t)$ is the total external force applied to the element per unit length, and the term on the right side of the equality is the inertial force of the element. The assumption of small shear deformation used in the force balance of equation (6.86) is true if $l/h_z \geq 10$ and $l/h_y \geq 10$ (i.e., for long slender beams or Euler–Bernoulli beams).

Next, the moments acting on the element dx about the z axis through point Q are summed. This yields

$$\left[M(x,t) + \frac{\partial M(x,t)}{\partial x} dx \right] - M(x,t) + \left[V(x,t) + \frac{\partial V(x,t)}{\partial x} dx \right] dx + [f(x,t)dx] \frac{dx}{2} = 0 \quad (6.87)$$

Here the right-hand side of the equation is zero since it is also assumed that the rotary inertia of the element dx is negligible. Simplifying this expression yields

$$\left[\frac{\partial M(x,t)}{\partial x} + V(x,t) \right] dx + \left[\frac{\partial V(x,t)}{\partial x} + \frac{f(x,t)}{2} \right] (dx)^2 = 0 \quad (6.88)$$

Since dx is assumed to be very small, $(dx)^2$ is assumed to be almost zero, so that this moment expression yields (dx is small, but not zero)

$$V(x,t) = -\frac{\partial M(x,t)}{\partial x} \quad (6.89)$$

This states that the shear force is proportional to the spatial change in the bending moment. Substitution of this expression for the shear force into equation (6.86) yields

$$-\frac{\partial^2}{\partial x^2}[M(x,t)]dx + f(x,t)dx = \rho A(x) dx \frac{\partial^2 w(x,t)}{\partial t^2} \quad (6.90)$$

Further substitution of equation (6.85) into (6.90) and dividing by dx yields

$$\rho A(x) \frac{\partial^2 w(x,t)}{\partial t^2} + \frac{\partial^2}{\partial x^2} \left[EI(x) \frac{\partial^2 w(x,t)}{\partial x^2} \right] = f(x,t) \quad (6.91)$$

If no external force is applied so that $f(x, t) = 0$ and if $EI(x)$ and $A(x)$ are assumed to be constant, equation (6.91) simplifies so that free vibration is governed by

$$\frac{\partial^2 w(x,t)}{\partial t^2} + c^2 \frac{\partial^4 w(x,t)}{\partial x^4} = 0, \quad c = \sqrt{\frac{EI}{\rho A}} \quad (6.92)$$

Note that unlike the previous equations, the free vibration equation (6.92) contains four spatial derivatives and hence requires four (instead of two) boundary conditions in calculating a solution. The presence of the two time derivatives again requires that two initial conditions, one for the displacement and one for the velocity, be specified.

The boundary conditions required to solve the spatial equation in a separation-of-variables solution of equation (6.92) are obtained by examining the deflection $w(x, t)$, the slope of the deflection $\partial w(x,t)/\partial x$, the bending moment $EI\partial^2 w(x,t)/\partial x^2$, and the shear force $\partial[EI\partial^2 w(x,t)/\partial x^2]/\partial x$ at each end of the beam. A common configuration is *clamped–free* or *cantilevered* as illustrated in Figure 6.10. In addition to a boundary being clamped or free, the end of a beam could be resting on a support restrained from bending or deflecting. The situation is called *simply supported* or *pinned*. A *sliding* boundary is one in which displacement is allowed but rotation is not. The shear load at a sliding boundary is zero.

If a beam in transverse vibration is free at one end, the deflection and slope at that end are unrestricted, but the bending moment and shear force must vanish:

$$\begin{aligned} \text{bending moment} &= EI\frac{\partial^2 w}{\partial x^2} = 0 \\ \text{shear force} &= \frac{\partial}{\partial x}\left[EI\frac{\partial^2 w}{\partial x^2}\right] = 0 \end{aligned} \tag{6.93}$$

If, on the other hand, the end of a beam is clamped (or fixed), the bending moment and shear force are unrestricted, but the deflection and slope must vanish at that end:

$$\begin{aligned} \text{deflection} &= w = 0 \\ \text{slope} &= \frac{\partial w}{\partial x} = 0 \end{aligned} \tag{6.94}$$

At a simply supported or pinned end, the slope and shear force are unrestricted and the deflection and bending moment must vanish:

$$\begin{aligned} \text{deflection} &= w = 0 \\ \text{bending moment} &= EI\frac{\partial^2 w}{\partial x^2} = 0 \end{aligned} \tag{6.95}$$

At a sliding end, the slope or rotation is zero and no shear force is allowed. On the other hand, the deflection and bending moment are unrestricted. Hence, at a sliding boundary,

$$\begin{aligned} \text{slope} &= \frac{\partial w}{\partial x} = 0 \\ \text{shear force} &= \frac{\partial}{\partial x}\left(EI\frac{\partial^2 w}{\partial x^2}\right) = 0 \end{aligned} \tag{6.96}$$

Other boundary conditions are possible by connecting the ends of a beam to a variety of devices such as lumped masses, springs, and so on. These boundary conditions can be determined by force and moment balances.

In addition to satisfying four boundary conditions, the solution of equation (6.92) for free vibration can be calculated only if two initial conditions (in time) are specified. As in the case of the rod, string, and bar, these initial conditions are specified initial deflection and velocity profiles:

$$w(x, 0) = w_0(x) \quad \text{and} \quad w_t(x, 0) = \dot{w}_0(x)$$

assuming that $t = 0$ is the initial time. Note that if w_0 and $\dot{w}_0$ were both zero, no motion would result.

The solution of equation (6.92) subject to four boundary conditions and two initial conditions proceeds following exactly the same steps used in previous sections. A separation-of-variables solution of the form $w(x, t) = X(x)T(t)$ is assumed. This is substituted into the equation of motion, equation (6.92), to yield (after rearrangement)

$$c^2 \frac{X''''(x)}{X(x)} = -\frac{\ddot{T}(t)}{T(t)} = \omega^2 \tag{6.97}$$

where the partial derivatives have been replaced with total derivatives as before (*note:* $X'''' = d^4X/dx^4$, $\ddot{T} = d^2T/dt^2$). Here the choice of separation constant, ω^2, is made, based on experience with the systems of Section 6.4 that the natural frequency comes from the temporal equation

$$\ddot{T}(t) + \omega^2 T(t) = 0 \tag{6.98}$$

which is the right side of equation (6.97). This temporal equation has a solution of the form

$$T(t) = A \sin \omega t + B \cos \omega t \tag{6.99}$$

where the constants A and B will eventually be determined by the specified initial conditions after being combined with the spatial solution.

The spatial equation comes from rearranging equation (6.97), which yields

$$X''''(x) - \left(\frac{\omega}{c}\right)^2 X(x) = 0 \tag{6.100}$$

By defining [recall equation (6.92)]

$$\beta^4 = \frac{\omega^2}{c^2} = \frac{\rho A \omega^2}{EI} \left(\text{so that } \omega = \beta^2 \sqrt{\frac{EI}{\rho A}} \text{ rad/s}\right) \tag{6.101}$$

and assuming a solution to equation (6.100) of the form $Ae^{\sigma x}$, the general solution of equation (6.100) can be calculated to be of the form (see Problem 6.44)

$$X(x) = a_1 \sin \beta x + a_2 \cos \beta x + a_3 \sinh \beta x + a_4 \cosh \beta x \tag{102}$$

Here the value for β and three of the four constants of integration a_1, a_2, a_3, and a_4 will be determined from the four boundary conditions. The fourth constant becomes combined with the constants A and B from the temporal equation, which are then determined from the initial conditions. The following example illustrates the solution procedure for a beam fixed at one end and simply supported at the other end.

Example 6.5.1

Calculate the natural frequencies and mode shapes for the transverse vibration of a beam of length l that is fixed at one end and pinned at the other end.

Solution The boundary conditions in this case are given by equation (6.94) at the fixed end and equation (6.95) at the pinned end. Substitution of the general solution given by equation (6.102) into equation (6.94) at $x = 0$ yields

$$X(0) = 0 \Rightarrow a_2 + a_4 = 0 \tag{a}$$

$$X'(0) = 0 \Rightarrow \beta(a_1 + a_3) = 0 \tag{b}$$

Similarly, at $x = l$ the boundary conditions result in

$$X(l) = 0 \Rightarrow a_1 \sin\beta l + a_2 \cos\beta l + a_3 \sinh\beta l + a_4 \cosh\beta l = 0 \tag{c}$$

$$EIX''(l) = 0 \Rightarrow \beta^2(-a_1 \sin\beta l - a_2 \cos\beta l + a_3 \sinh\beta l + a_4 \cosh\beta l) = 0 \tag{d}$$

These four boundary conditions thus yield four equations [(a) through (d)] in the four unknown coefficients a_1, a_2, a_3, and a_4. These can be written as the single vector equation

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ \beta & 0 & \beta & 0 \\ \sin\beta l & \cos\beta l & \sinh\beta l & \cosh\beta l \\ -\beta^2 \sin\beta l & -\beta^2 \cos\beta l & \beta^2 \sin\beta l & \beta^2 \cosh\beta l \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Recall from Chapter 4 that this vector equation can have a nonzero solution for the vector $\mathbf{a} = [a_1 \quad a_2 \quad a_3 \quad a_4]^T$ only if the determinant of the coefficient matrix vanishes (i.e., if the coefficient matrix is singular). Furthermore, recall that since the coefficient matrix is singular, not all of the elements of the vector $\mathbf{a}$ can be calculated.

Setting the determinant above equal to zero yields the characteristic equation

$$\tan\beta l = \tanh\beta l$$

This equality is satisfied for an infinite number of choices for β, denoted β_n. The solution can be visualized by plotting both $\tan\beta l$ and $\tanh\beta l$ versus βl on the same plot. This is similar to the solution technique used in Example 6.4.1 and illustrated in Figure 6.9. The first five solutions are

$$\beta_1 l = 3.926602 \qquad \beta_2 l = 7.068583 \qquad \beta_3 l = 10.210176$$

$$\beta_4 l = 13.351768 \qquad \beta_5 l = 16.49336143$$

For the rest of the modes (i.e., for values of the index $n > 5$), the solutions to the characteristic equation are well approximated by

$$\beta_n l = \frac{(4n + 1)\pi}{4}$$

The weighted frequencies determine the system natural frequencies by

$$\omega_n = \beta_n^2 \sqrt{\frac{EI}{\rho A}}\ \text{rad/s}, \quad \text{and} \quad f_n = \frac{\beta_n^2}{2\pi}\sqrt{\frac{EI}{\rho A}}\ \text{Hz}$$

With these values of the weighted frequencies $\beta_n l$, the individual modes of vibration can be calculated. Solving the preceding matrix equation for the individual coefficients a_i yields $a_1 = -a_3$, $a_2 = -a_4$, and

$$(\sinh \beta_n l - \sin \beta_n l)a_3 + (\cosh \beta_n l - \cos \beta_n l)a_4 = 0$$

Thus

$$a_3 = -\frac{\cosh \beta_n l - \cos \beta_n l}{\sinh \beta_n l - \sin \beta_n l} a_4$$

for each n. The fourth coefficient a_4 cannot be determined by this set of equations, because the coefficient matrix is singular (otherwise, each a_i would be zero). This remaining coefficient becomes the arbitrary magnitude of the eigenfunction. As this constant depends on n, denote it by $(a_4)_n$. Substitution of these values of a_i in the expression $X(x)$ for the spatial solution yields the result that the eigenfunctions or mode shapes have the form

$$X_n(x) = (a_4)_n \left[\frac{\cosh \beta_n l - \cos \beta_n l}{\sinh \beta_n l - \sin \beta_n l}(\sinh \beta_n x - \sin \beta_n x) - \cosh \beta_n x + \cos \beta_n x \right],$$
$$n = 1, 2, 3, \ldots$$

The first three mode shapes are plotted in Figure 6.11 for $(a_2)_n = 1$ and $n = 1, 2, 3$.

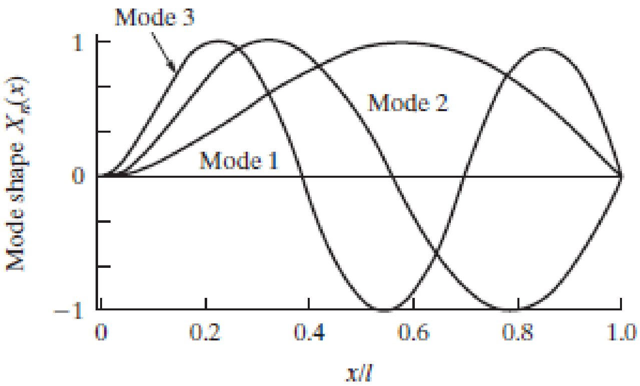


Figure 6.11 A plot of the first three mode shapes of the clamped–pinned beam of Example 6.5.1, arbitrarily normalized to unity.

These mode shapes can be shown to be orthogonal, so that

$$\int_0^l X_n(x)X_m(x)dx = 0$$

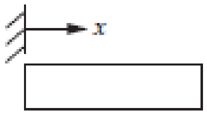
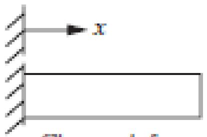
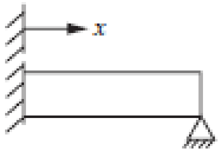
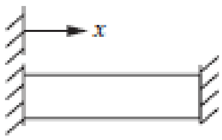
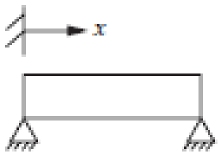
for $n \neq m$ (see Problem 6.47). As in Example 6.3.2, this orthogonality, along with the initial conditions, can be used to calculate the constants A_n and B_n in the series solution for the displacement

$$w(x, t) = \sum_{n=1}^{\infty}(A_n \sin \omega_n t + B_n \cos \omega_n t)X_n(x)$$

□

Table 6.6 summarizes a number of different boundary configurations for the slender beam. The slender-beam model given in equation (6.91) is often referred to

TABLE 6.6 SAMPLE OF VARIOUS BOUNDARY CONFIGURATIONS OF A SLENDER BEAM IN TRANSVERSE VIBRATION OF LENGTH l ILLUSTRATING WEIGHTED NATURAL FREQUENCIES AND MODE SHAPES[a]

Configuration	Weighted frequencies $\beta_n l$ and characteristic equation	Mode shape	σ_n
	0 (rigid-body mode)	$\cosh\beta_n x + \cos\beta_n x$	
	4.73004074		0.9825
	7.85320462	$-\sigma_n(\sinh\beta_n x + \sin\beta_n x)$[b]	1.0008
	10.9956078		0.9999
	14.1371655		1.0000
Free–free	17.2787597		0.9999
	$\frac{(2n+1)\pi}{2}$ for $n > 5$		1 for $n > 5$
	$\cos\beta l \cosh\beta l = 1$		
	1.87510407	$\cosh\beta_n x - \cos\beta_n x$	0.7341
	4.69409113		1.0185
	7.85475744	$-\sigma_n(\sinh\beta_n x - \sin\beta_n x)$	0.9992
	10.99554073		1.0000
Clamped–free	14.13716839		1.0000
	$\frac{(2n-1)\pi}{2}$ for $n > 5$		1 for $n > 5$
	$\cos\beta l \cosh\beta l = -1$		
	3.92660231	$\cosh\beta_n x - \cos\beta_n x$	1.0008
	7.06858275		1 for $n > 1$
	10.21017612	$-\sigma_n(\sinh\beta_n x - \sin\beta_n x)$	
	13.35176878		
Clamped–pinned	16.49336143		
	$\frac{(4n+1)\pi}{4}$ for $n > 5$		
	$\tan\beta l = \tanh\beta l$		
	2.36502037	$\cosh\beta_n x - \cos\beta_n x$	0.9825
	5.49780392		1 for $n > 1$
	8.63937983	$-\sigma_n(\sinh\beta_n x - \sin\beta_n x)$	
	11.78097245		
	14.92256510		
Clamped–sliding	$\frac{(4n-1)\pi}{4}$ for $n > 5$		
	$\tan\beta l + \tanh\beta l = 0$		
	4.73004074	$\cosh\beta_n x - \cos\beta_n x$	0.982502
	7.85320462		1.00078
	10.9956079	$-\sigma_n(\sinh\beta_n x - \sin\beta_n x)$	0.999966
	14.1371655		1.0000
	17.2787597		1.0000
Clamped–clamped	$\frac{(2n+1)\pi}{2}$ for $n > 5$		1 for $n > 5$
	$\cos\beta l \cosh\beta l = 1$		
	$n\pi$	$\sin\frac{n\pi x}{l}$	none
	$\sin\beta l = 0$		
Pinned–pinned			

[a]Here the weighted natural frequencies $\beta_n l$ are related to the natural frequencies by equation (6.101) or $\omega_n = \beta_n^2\sqrt{EI/\rho A}$, as used in Example 6.5.1. The values of σ_i for the mode shapes are computed from the formulas given in Table 6.5.

[b]There are two free–free mode shapes: X_0 = constant and $X_0 = A(x - l/2)$; the first is translational, the second rotational.

as the *Euler–Bernoulli* or Bernoulli–Euler beam equation. The assumptions used in formulating this model are that the beam be

- Uniform along its span, or length, and slender ($l > 10h$)
- Composed of a linear, homogeneous, isotropic elastic material without axial loads
- Such that plane sections remain plane
- Such that the plane of symmetry of the beam is also the plane of vibration so that rotation and translation are decoupled
- Such that rotary inertia and shear deformation can be neglected

The key to solving for the time response of distributed-parameter systems is the orthogonality of the mode shapes. Note from Table 6.7 that the mode shapes are quite complicated in many configurations. This does not mean that orthogonality is necessarily violated, just that evaluating the integrals in the modal analysis procedure becomes more difficult.

TABLE 6.7 EQUATIONS FOR THE MODE SHAPE COEFFICIENTS σ_n FOR USE IN TABLE 6.4[a]

Boundary condition	Formula for σ_n
Free–free	$\sigma_n = \dfrac{\cosh \beta_n l - \cos \beta_n l}{\sinh \beta_n l - \sin \beta_n l}$
Clamped–free	$\sigma_n = \dfrac{\sinh \beta_n l - \sin \beta_n l}{\cosh \beta_n l + \cos \beta_n l}$
Clamped–pinned	$\sigma_n = \dfrac{\cosh \beta_n l - \cos \beta_n l}{\sinh \beta_n l - \sin \beta_n l}$
Clamped–sliding	$\sigma_n = \dfrac{\sinh \beta_n l - \sin \beta_n l}{\cosh \beta_n l + \cos \beta_n l}$
Clamped–clamped	Same as free–free

[a]These coefficients are used in the calculations for the mode shapes, as illustrated in Example 6.5.1.

Livro do Blevins

Table 8-1. Single-Span Beams.

Notation: x = distance along span of beam; m = mass per unit length of beam; $\bar{m} = \frac{m}{L}$
E = modulus of elasticity;
I = area moment of inertia of beam about neutral axis (Table 5-1); L = span of beam;
see Table 3-1 for consistent sets of units

Natural Frequency (hertz); $f_i = \frac{\lambda_i^2}{2\pi L^2}\left(\frac{EI}{m}\right)^{1/2}$; i=1,2,3... $f = \frac{\lambda^2}{2\pi L^2}\cdot\left(\frac{Eh^2}{12\rho}\right)^{1/2}$

Description(a)	λ_i; i=1,2,3...	Mode Shape, $\tilde{y}_i\left(\frac{x}{L}\right)$	σ_i; i=1,2,3...
1. Free-Free	4.73004074 7.85320462 10.9956078 14.1371655 17.2787597 $(2i+1)\frac{\pi}{2}$; i>5	$\cosh\frac{\lambda_i x}{L} + \cos\frac{\lambda_i x}{L} - \sigma_i\left(\sinh\frac{\lambda_i x}{L} + \sin\frac{\lambda_i x}{L}\right)$	0.982502215 1.000777312 0.999966450 1.000001450 0.999999937 ≈1.0 for i>5 See Ref. 8-2
2. Free-Sliding	2.36502037 5.49780392 8.63937983 11.78097245 14.92256510 $(4i-1)\frac{\pi}{4}$; i>5	$\cosh\frac{\lambda_i x}{L} + \cos\frac{\lambda_i x}{L} - \sigma_i\left(\sinh\frac{\lambda_i x}{L} + \sin\frac{\lambda_i x}{L}\right)$	0.982502207 0.999966450 0.999999933 0.999999993 0.999999993 1.0; i>5
3. Clamped-Free	1.87510407 4.69409113 7.85475744 10.99554073 14.13716839 $(2i-1)\frac{\pi}{2}$; i>5	$\cosh\frac{\lambda_i x}{L} - \cos\frac{\lambda_i x}{L} - \sigma_i\left(\sinh\frac{\lambda_i x}{L} - \sin\frac{\lambda_i x}{L}\right)$	0.734095514 1.018467319 0.999224497 1.000033553 0.999998550 ≈1.0; i>5 See Ref. 8-2
			1.000777304

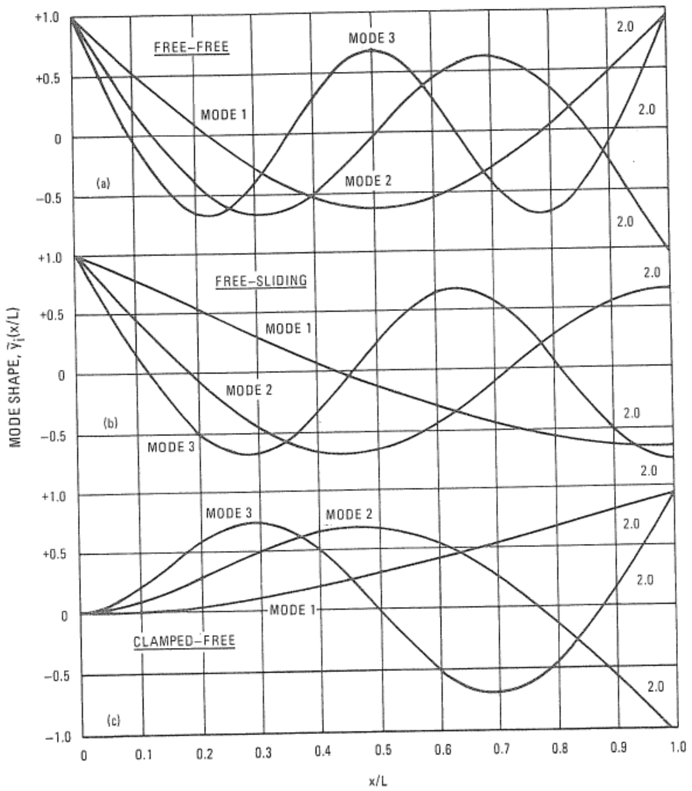


Fig. 8-4. Normalized mode shapes of first three modes of straight slender beams with the various boundary conditions corresponding to Table 8-1. These mode shapes have been divided by the normalization constants given in the graphs.

2.5 HYSTERETIC DAMPING - GENERAL CASE

The analysis in the previous section for proportionally-damped systems gives some insight into the characteristics of this more general description of practical structures. However, as was stated there, the case of proportional damping is a particular one which, although often justified in a theoretical analysis because it is realistic and also because of a lack of any more accurate model, does not apply to all cases. In our studies here, it is very important that we consider the most general case if we are to be able to interpret and analyse correctly the data we observe on real structures. These, after all, know nothing of our predilection for assuming proportionality. Thus, we consider in the next two sections the properties of systems with general damping elements, first of the hysteretic type, then viscous.

We start by writing the general equation of motion for a MDOF system with hysteretic damping and harmonic excitation (as it is this that we are working towards):

$$[M]\{\ddot{x}\} + [K]\{x\} + i[H]\{x\} = \{f\}e^{i\omega t} \qquad (2.56)$$

Now, consider first the case where there is no excitation and assume a solution of the form:

$$\{x\} = \{X\}e^{i\lambda t} \qquad (2.57)$$

Substituted into (2.56), this trial solution leads to a complex eigenproblem whose solution is in the form of two matrices (as for the earlier undamped case), containing the eigenvalues and eigenvectors. In this case, however, these matrices are both complex, meaning that each natural frequency and each mode shape is described in terms of complex quantities. We choose to write the r^{th} eigenvalue as

$$\lambda_r^2 = \omega_r^2(1+i\eta_r) \qquad (2.58)$$

where ω_r is the natural frequency and η_r is the damping loss factor for that mode. It is important to note that the natural frequency ω_r is not (necessarily) equal to the natural frequency of the undamped system, $\bar{\omega}_r$, as was the case for proportional damping, although the two values will generally be very close in practice.

The complex mode shapes are at first more difficult to interpret but in fact what we find is that the amplitude of each coordinate has both a magnitude and a phase angle. This is only very slightly different from the undamped case as there we effectively have a magnitude at each point plus a phase angle which is either 0° or 180°, both of which can be completely described using real numbers. What the inclusion of general damping effects does is to generalise this particular feature of the mode shape data.

This eigensolution can be seen to possess the same type of orthogonality properties as those demonstrated earlier for the undamped system and may be defined by the equations:

$$[\Psi]^T[M][\Psi] = [\,m_r\,]$$

$$[\Psi]^T\,[K+iH]\,[\Psi] = [\,k_r\,] \qquad (2.59)$$

Again, the generalised mass and stiffness parameters (now complex) depend upon the normalisation of the mode shape vectors for their magnitudes but always obey the relationship:

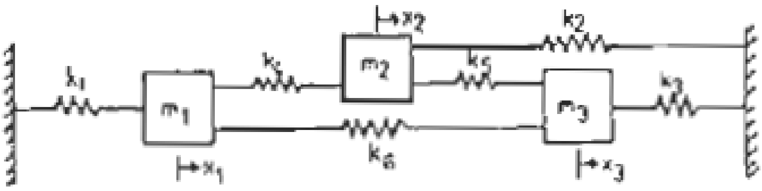
$$\lambda_r^2 = k_r/m_r \qquad (2.60)$$

and here again we may define a set of mass-normalised eigenvectors as:

$$\{\phi\}_r = (m_r)^{-1/2}\,\{\psi\}_r \qquad (2.61)$$

NUMERICAL EXAMPLES WITH STRUCTURAL DAMPING

Some further numerical examples are included to illustrate the characteristics of more general damped systems, based on the following 3DOF model:



MODEL 1

$m_1 = 0.5$ kg $\qquad k_1 = k_2 = k_3 = k_4 = k_5 = k_6$

$m_2 = 1.0$ kg $\qquad = 1.0 \times 10^3$ N/m

$m_3 = 1.5$ kg

Case 1(a) – Undamped

$$[\ \omega_r^2\ ] = \begin{bmatrix} 950 & 0 & 0 \\ 0 & 3352 & 0 \\ 0 & 0 & 6698 \end{bmatrix} ; [\Phi] = \begin{bmatrix} 0.464 & -0.218 & -1.318 \\ 0.536 & -0.782 & 0.318 \\ 0.635 & 0.493 & 0.142 \end{bmatrix}$$

Case 1(b) – Proportional Structural Damping ($h_j = 0.05k_j$ $j=1,6$)

$$[\ \lambda_r^2\ ] = \begin{bmatrix} 950(1+.05i) & 0 & 0 \\ 0 & 3352(1+0.05i) & 0 \\ 0 & 0 & 6698(1+.05i) \end{bmatrix}$$

$$[\ \Phi\ ] = \begin{bmatrix} 0.464(0^\circ) & 0.218(180^\circ) & 1.318(180^\circ) \\ 0.536(0^\circ) & 0.782(180^\circ) & 0.318(0^\circ) \\ 0.635(0^\circ) & 0.493(0^\circ) & 0.142(0^\circ) \end{bmatrix}$$

Case 1(c) – Non-Proportional Structural Damping

($h_1 = 0.1k_1$, $h_{2-6} = 0$: ie a single damper between m_1 and ground)

$$[\ \lambda_r^2\ ] = \begin{bmatrix} 957(1+.0671i) & 0 & 0 \\ 0 & 3354(1+.00421i) & 0 \\ 0 & 0 & 6690(1+.0781i) \end{bmatrix}$$

$$[\ \Phi\ ] = \begin{bmatrix} 0.463(-5.5^\circ) & 0.217(173^\circ) & 1.321(181^\circ) \\ 0.537(0^\circ) & 0.784(181^\circ) & 0.316(-6.7^\circ) \\ 0.636(1.0^\circ) & 0.492(-1.3^\circ) & 0.142(-3.1^\circ) \end{bmatrix}$$

NOTES: (i) Each mode has a different damping factor

(ii) All eigenvector arguments within 10^{o} of 0^{o} or 180^{o} (ie the modes are almost 'real').

MODEL 2

$m_1 = 1.0$ kg
$m_2 = 0.95$ kg
$m_3 = 1.05$ kg

$k_1 = k_2 = k_3 = k_4 = k_5 = k_6$
$= 1 \times 10^3$ N/m

Case 2(a) - Undamped

$$\lceil \omega_r^2 \rfloor = \begin{bmatrix} 999 & 0 & 0 \\ 0 & 3892 & 0 \\ 0 & 0 & 4124 \end{bmatrix}, \quad [\Phi] = \begin{bmatrix} 0.577 & -0.702 & 0.552 \\ 0.567 & -0.215 & -0.826 \\ 0.587 & 0.752 & 0.207 \end{bmatrix}$$

NOTE: This system has two close natural frequencies

Case 2(b) - Proportional Structural Damping ($h_j = 0.05k_j$)

$$\lceil \lambda_r^2 \rfloor = \begin{bmatrix} 999(1+.05i) & 0 & 0 \\ 0 & 3992(1+.05i) & 0 \\ 0 & 0 & 4124(1+.05i) \end{bmatrix}$$

$$[\Phi] = \begin{bmatrix} 0.577(0^{o}) & 0.702(180^{o}) & 0.552(0^{o}) \\ 0.567(0^{o}) & 0.215(180^{o}) & 0.827(180^{o}) \\ 0.587(0^{o}) & 0.752(0^{o}) & 0.207(0^{o}) \end{bmatrix}$$

Case 2(c) - Non-Proportional Structural Damping
($h_1 = 0.1k_1$; $h_{2-6}=0$)

$$\lceil \lambda_r^2 \rfloor = \begin{bmatrix} 1006(1+.10i) & 0 & 0 \\ 0 & 3942(1+.031i) & 0 \\ 0 & 0 & 4067(1+.019i) \end{bmatrix}$$

$$[\Phi] = \begin{bmatrix} 0.578(-4^{o}) & 0.851(162^{o}) & 0.685(40^{o}) \\ 0.569(2^{o}) & 0.570(101^{o}) & 1.019(176^{o}) \\ 0.589(2^{o}) & 0.840(12^{o}) & 0.560(-50^{o}) \end{bmatrix}$$

Forced Response Analysis

We turn next to the analysis of forced vibration for the particular case of harmonic excitation and response, for which the governing equation of motion is:

$$[K + iH - \omega^2 M]\{x\}e^{i\omega t} = \{f\}e^{i\omega t} \qquad (2.62)$$

As before, a direct solution to this problem may be obtained by using the equations of motion to give:

$$\{x\} = [K + iH - \omega^2 M]^{-1}\{f\} = [\alpha(\omega)]\{f\} \qquad (2.63)$$

but again this is very inefficient for numerical application and we shall make use of the same procedure as before by multiplying both sides of the equation by the eigenvectors. Starting with (2.63), and following the same procedure as between equations (2.37) and (2.39), we can write:

$$[\alpha(\omega)] = [\Phi]\,[\lambda_r^2 - \omega^2]^{-1}\,[\Phi]^T \qquad (2.64)$$

and from this full matrix equation we can extract any one FRF element, such as $\alpha_{jk}(\omega)$ and express it explicitly in a series form:

$$\alpha_{jk}(\omega) = \sum_{r=1}^{N} \frac{{}_r\phi_j\,{}_r\phi_k}{\omega_r^2-\omega^2+i\eta_r\omega_r^2} \qquad (2.65)$$

which may also be rewritten in various alternative ways, such as:

$$\alpha_{jk}(\omega) = \sum_{r=1}^{N} \frac{{}_r\psi_j\,{}_r\psi_k}{m_r(\omega_r^2-\omega^2+i\eta_r\omega_r^2)}$$

or

$$= \sum_{r=1}^{N} \frac{{}_rA_{jk}}{\omega_r^2-\omega^2+i\eta_r\omega_r^2}$$

In these expressions, the numerator (as well as the denominator) is now complex as a result of the complexity of the eigenvectors. It is in this respect that the general damping case differs from that for proportional damping.